

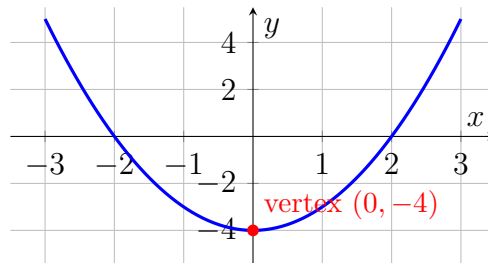
Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____
8. _____ 9. _____

Quadratic Functions: Vertex, Max & Min

Example (State max or min, the vertex, and the intervals):

For $y = x^2 - 4$ the vertex is $(0, -4)$, a **minimum**. It decreases for $x < 0$ and increases for $x > 0$.



Directions: For each function state max or min, give the vertex, and the increasing / decreasing intervals.

1. $y = x^2 + 2$

4. $y = -(x + 1)^2 - 2$

2. $y = -x^2 + 5$

5. $y = (x + 2)^2 - 1$

3. $y = (x - 3)^2$

6. $y = -(x - 4)^2 + 3$

7. $y = x^2 - 6x$

8. $y = 2(x - 1)^2 + 3$

Challenge (same sheet):

9. $y = (x - 2)^2 - 9$ — give the vertex, both intervals, and the two x -intercepts.

Answer Key — fully worked

Procedure: Opens up ($+x^2$) gives a minimum; opens down gives a maximum; Vertex form $a(x - h)^2 + k$ has vertex (h, k) ; The graph turns (changes direction) at the vertex.

1. Opens up $\Rightarrow$ min; vertex $(0, 2)$; dec $x < 0$, inc $x > 0$
2. Opens down $\Rightarrow$ max; vertex $(0, 5)$; inc $x < 0$, dec $x > 0$
3. Vertex $(3, 0)$, opens up $\Rightarrow$ min; dec $x < 3$, inc $x > 3$
4. Vertex $(-1, -2)$, opens down $\Rightarrow$ max; inc $x < -1$, dec $x > -1$
5. Vertex $(-2, -1)$, opens up $\Rightarrow$ min; dec $x < -2$, inc $x > -2$
6. Vertex $(4, 3)$, opens down $\Rightarrow$ max; inc $x < 4$, dec $x > 4$
7. Complete the square: $x^2 - 6x = (x - 3)^2 - 9$; vertex $(3, -9)$ min; dec $x < 3$, inc $x > 3$
8. $a = 2 > 0$ opens up $\Rightarrow$ min; vertex $(1, 3)$; dec $x < 1$, inc $x > 1$
9. Vertex $(2, -9)$ min; dec $x < 2$, inc $x > 2$; intercepts $x = -1, 5$